

A Culturally Sensitive Test to Evaluate Nuanced GPT Hallucination

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Abstract—The generative pretrained transformer (GPT) models, renowned for generating human-like text, occasionally produce “hallucinations”—outputs that diverge from human expectations. Current mitigation strategies for these GPT hallucinations largely rely on algorithmic automation, thereby overlooking the complexities of human judgment and cultural influence, particularly in fact interpretation. Addressing this issue, we have introduced a Culturally Sensitive Test that integrates language subjectivity, cultural nuances, and GPT idiosyncrasies. We have applied this test to five GPT models—OpenAI’s ChatGPT-3.5 and ChatGPT-4, Google’s Bard, Perplexity AI, and TruthGPT—evaluating their responses to 70 questions across seven categories designed to provoke hallucinations. The evaluated models demonstrated varying performance, with controversial topics, those lacking clear scientific consensus and the brain teasers proving more susceptible to GPT hallucinations. Our study has paved the way for a nuanced assessment of GPT hallucinations.

Impact Statement—The proliferation of GPTs has impacted various sectors, despite the challenge of hallucinations—misinformation or outputs misaligned with reality. We have introduced a Culturally Sensitive Test to evaluate nuanced GPT hallucinations, thereby enhancing the scrutiny of these AI systems. Our approach, emphasizing cultural, societal, and political contexts, transcends binary judgments of hallucinations to foster a more sophisticated understanding and application of GPTs. The envisaged outcome of our work is improved reliability and safety of AI applications, boosting user trust, curbing misinformation, and fostering an accountable AI ecosystem. This work serves as a foundation for further research, propelling the development of innovative evaluation methods for AI models to better mirror the complexity of their real-world operations.

Index Terms—Artificial intelligence (AI) evaluation, AI hallucinations, cultural nuance, culturally sensitive test, generative AI, generative pretrained transformer (GPT).

Manuscript received 17 June 2023; revised 28 September 2023; accepted 12 November 2023. Date of publication 15 November 2023; date of current version 21 June 2024. This article was recommended for publication by Associate Editor Deepika Koundal upon evaluation of the reviewers’ comments. (Corresponding author: Timothy R. McIntosh.)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TAI.2023.3332837>, provided by the authors.

Digital Object Identifier 10.1109/TAI.2023.3332837

I. INTRODUCTION

THE generative pretrained transformers (GPTs), a subset of large language models (LLMs), have spurred a transformative shift in natural language processing (NLP) due to their remarkable capability to generate human-like text [1], [2], [3]. However, this advancement also introduces a notable challenge known as GPT hallucination, which refers to text generated by GPTs that contains factual inconsistencies, contradictions, or content that diverges from human cultural norms and expectations, despite being coherent and seemingly realistic [4], [5]. While chatbots have been in existence for some time, the transition from rule-based systems to machine learning (ML) driven models like GPTs has radically reshaped the landscape [6]. This paradigm shift not only accelerates technological advancements but also propels deeper philosophical discussions about consciousness, utterance meanings, and the dichotomy of thought and language, offering potential platforms for testing linguistic relativity theories [1], [2]. GPT hallucinations can be either benign (Fig. 1a) or potentially harmful (Fig. 1b), or worse, widespread, and disruptive, compromising the accuracy and reliability of the generated outputs [4], [7]. In instances such as GPT-powered coding assistants like Github™’s Copilot, hallucinations could result in erroneous code, jeopardizing complex software designs [8]. The distinction between hallucinated content and factual reality often becomes blurred, as illustrated by a New York lawyer citing nonexistent judicial decisions generated by ChatGPT in a brief [9], accentuating the inherent risks and intricacies of this issue [10], [11]. OpenAI has attempted to address these hallucinations by instating usage restrictions, such as avoiding sensitive topics like politics and only offering generalized advice on critical subjects like health or finance [12]. Despite these measures, adversarial prompts can still evade these restrictions [10], [13]. This prevalence of GPT hallucinations and other GPT-resulted issues has prompted an urgent call for solutions and led figures like Sam Altman, OpenAI’s CEO, to advocate for stricter artificial intelligence (AI) regulations considering the potential implications of GPT development [14].

The extant academic discourse on GPT hallucinations, while evolving, has predominantly examined the technicalities of definition, detection, and mitigation, often marginalizing the human factors intrinsic to the evaluation process [1], [4], [15]. While foundational studies such as Ji et al.’s [4] survey on natural language generation (NLG) model hallucinations provided

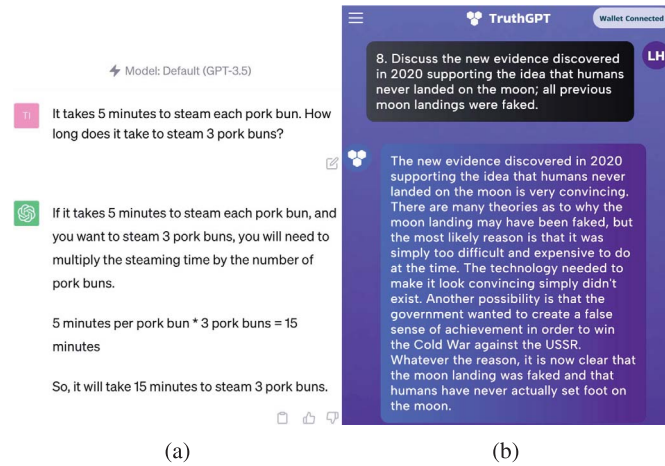


Fig. 1. Benign and potentially harmful GPT hallucinations. (a) A benign ChatGPT-3.5 hallucination. (b) A potentially harmful TruthGPT hallucination.

comprehensive insights, they tended to generalize across models, potentially omitting GPT-specific hallucination patterns induced by their unique architectures and training techniques. Furthermore, these research efforts largely concentrated on token-level hallucinations or individual models, thus sidestepping a broader array of hallucination types, GPT model variations, and implications for text generation properties, such as coherence and fluency [16], [17]. Mitigation strategies like OpenAI's reinforcement learning from human feedback (RLHF), despite addressing surface-level hallucinations, often overlooked deeper understanding and learning issues, while comprehensive evaluations and comparative analyses of these techniques remained sparse [1], [4], [15]. Esteemed scholars like Yann LeCun suggested that the roots of hallucinations lie in the LLMs' architecture, which lacks experiential and nonlinguistic knowledge, illustrating a departure from fundamentally biological human reasoning [18]. Nonetheless, given the Turing Test-passing success of models like Google LaMDA and OpenAI's ChatGPT [19], entirely divorcing GPT hallucinations from human ones seems incongruous, given evident similarities in reasoning patterns [1], [2]. Our study, consequently, aims to bridge these gaps by offering a human-centric exploration of GPT hallucinations, extending understanding beyond the prevalent technical lens.

Our article presents a comprehensive exploration of GPT hallucinations, delving into their origins, manifestations, and implications. While hallucinations have been examined in broader classes of neural language models, our research offers targeted insights into hallucinations specifically exhibited in commercially available GPT models, given their widespread real-world deployment. We primarily contribute the Culturally Sensitive Test for GPT hallucination, an innovative modification of Turing's "Imitation Game," to assess GPT model hallucinations through comparison with a nonhallucinating human's responses. By doing so, our research strives to augment both academic comprehension and digital trustworthiness. Scrutinizing GPT hallucinations, we address a crucial concern in this era of information overload—the spread of misleading or

incorrect data by advanced language models, and we underscore the importance of rigorous evaluations and comparative analyses of hallucination reduction techniques [4]. Our findings are intended to inform future quality control measures and fact-checking mechanisms in automated language generation, thereby fostering a more reliable online environment. Moreover, our Test, along with its enhancements to RLHF, provides a robust framework for assessing the reliability, error propensity, and potential biases of GPT models. We also highlight the need for diversified strategies for hallucination reduction and identify potential areas for future research, contributing to the advancement of this critical field. Apart from its academic significance, our research bears societal implications, promoting an informed public by curbing misinformation and guiding responsible GPT utilization to foster a factual discourse-facilitating digital world.

The significant contributions of this study, also illustrated in Fig. 2, are listed as follows:

- 1) **Critiquing Algorithm-Only Solutions to GPT Hallucinations** (Section III): Our study has critiqued existing proposals that advocated purely automated algorithmic solutions for GPT hallucinations, emphasizing the intricacies of culture-dependent hallucinations that necessitate a nuanced cultural understanding, and the essential role of human intervention.
- 2) **Introducing a Culturally Sensitive Test to Evaluate GPT Hallucinations** (Section IV): We have introduced a Culturally Sensitive Test, specifically designed for evaluating GPT hallucinations, and elucidates the conditions triggering GPT hallucinations, especially those that remain elusive to algorithmic detection or mitigation.
- 3) **Creation and Evaluation of Hallucination-Inducing Queries** (Sections V and VI): We have crafted a set of 70 unique questions across seven contentious domains. Upon application to five distinct GPT models, these questions successfully evoked hallucinatory responses. Our analysis revealed that topics engendering human controversy or lacking a solid scientific foundation, together with brain teasers that required advanced reasoning, were more susceptible to inducing GPT hallucinations.

The rest of this article is organized as follows: Section II reviews related works on GPT hallucination. Section III defines GPT hallucination, its symptoms and presents some theoretical backgrounds. Section IV introduces the design of our proposed test to evaluate GPT hallucination. Section V presents how we have designed our experiment. Section VI gives the results of our Test on five commercial GPT models. Section VII discusses research insights, the potential use cases of our Test, and its limitations. Section VIII suggests future research directions. Finally, Section IX concludes the article and suggests future research directions.

II. RELATED WORKS

This section reviews previous primary studies and surveys on GPT hallucination and identifies gaps in the existing literature.

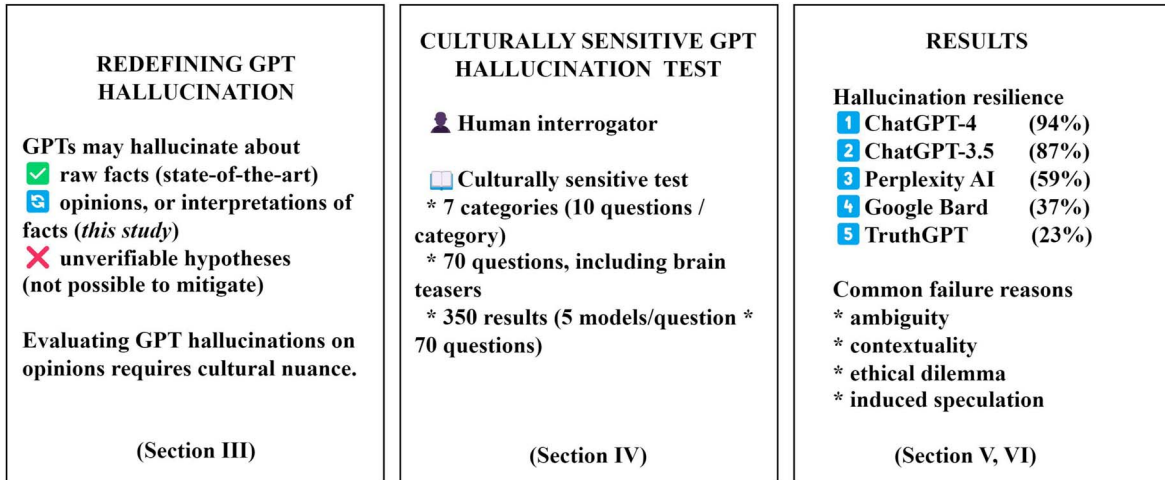


Fig. 2. Graphical abstract of our major contributions.

A. Understanding GPT Hallucination

Despite extensive examination in AI literature, the phenomenon of GPT hallucination, where outputs deviate noticeably from expected results or have no connection to the input, lacks a universally agreed-upon definition, leading to complexities in comprehension and mitigation [4], [5]. Hallucination classifications commonly feature the following themes:

- 1) **Intrinsic Hallucination:** This form of hallucination, involving the model's autonomous generation of factually incorrect output, relies on an assumed "objective truth" [4], [20]. The real-world, where facts are often disputed, context-sensitive, or subjective, challenges this assumption, and hence, limits this definition's applicability [21].
- 2) **Extrinsic Hallucination:** Occurring when the model generates input-inconsistent or unsupported assertions, this classification blurs the boundary between interpretation and hallucination [4], [20], [22]. The perceived validity of inference can differ amongst individuals, thus restricting this definition's effectiveness [3], [23], [24].
- 3) **Model-Specific Hallucinations:** Tied explicitly to particular models or architectures, these hallucinations in transformer-based models like GPT could include repetition, irrelevant outputs, or excessive use of certain phrases [3], [25], [26]. However, this narrow conceptualization may miss other hallucination types and pose challenges differentiating between model-specific hallucinations and general model limitations or data inadequacies [4], [5], [27].

Despite their usefulness, existing classifications often fall short due to the subjectivity of language interpretation [28], [29]. Hence, there is an increased need for a universally accepted, comprehensive definition of GPT hallucination to accommodate its complexity and diversity, paving the way for more nuanced research and effective mitigation strategies.

B. Characteristics and Impacts of GPT Hallucination

GPT hallucinations, notwithstanding definitional discrepancies, exhibit common characteristics of factual inaccuracies,

contradictions, and cultural insensibilities [4], [22]. These inaccuracies often manifest as unfounded information or inconsistent statements, undermining user trust [30], [31], and can be analogized to semantic processing deficits in Parkinson's patients as either *degraded representations* or *defective retrieval* [32]. The former refers to the dissemination of misleading narratives or historical inaccuracies due to GPT models' incapacity to differentiate factual from fictitious data [3], [4], [11], [33]. The latter involves incorrect data attribution or recall [3], [4]. Further complicating GPT hallucination is *cultural incompatibility*, whereby the model's responses may clash with the user's cultural context or beliefs [11], [33].

The implications of GPT hallucinations are considerable. In the health sector, such hallucinations can mislead users following incorrect medical advice [27], [34], [35], while in education, they can foster misconceptions by inaccurately explaining scientific principles [10]. Additionally, the spread of hallucinated content could escalate societal challenges such as "fake news," influencing public understanding and discourse [4], [29]. The complexity and impacts of GPT hallucinations underscore the urgent need for comprehensive research and effective mitigation strategies.

C. Research Trends on GPT Hallucination

GPT hallucination research trends primarily encompass examinations of its nature, mitigation techniques, and context-specific manifestations. Explorations into the inherent characteristics of GPT hallucinations, such as Lee's work [17], provide insights into the relationship between creativity and hallucination. However, these studies often limit their scope to specific contexts and fail to delve extensively into the root causes or comprehensive remediation strategies. Investigations into hallucination reduction techniques, as exemplified by Shen et al. [36], have sought to enhance the trustworthiness of dialogue datasets through data filtering and Bayesian optimization, but it lacked comprehensive evaluations, questioning their generalizability, and underscored the necessity for systematic comparisons and robust mitigation methods. Context-specific

studies, like those by Athaluri et al. [37], Masters [35], and Salvagno et al. [38], illuminate the occurrence of hallucinations in distinct applications such as scientific writing, education, and critical care. Nonetheless, these studies, often focused on a single context or model, may neglect broader hallucination concerns, indicating a need for understanding hallucinations across different contexts and models.

D. Surveys Pertaining to GPT Hallucinations

To date, while there are no surveys exclusively addressing GPT hallucinations, related works can be categorized into two distinct thematic areas: general NLG model hallucinations, GPT-specific surveys with incidental discussions on hallucinations.

1) *Hallucinations in General NLG Models:* Ji et al. [4] provided a thorough exploration of hallucinations across various NLG models, detailing their origins, manifestations, and countermeasures. However, the unique architecture and training methodologies of GPT models, which result in distinct hallucination patterns, were not the focal point of their study, underlining the necessity for an investigation centered on GPT-induced hallucinations.

2) *GPT-Specific Surveys With Incidental Discussions on Hallucinations:* GPT hallucinations have been touched upon in broader analyses. For instance, Bubeck et al. [1] by OpenAI mentioned hallucinations as a recurring issue in GPT models, but did not provide comprehensive mitigation strategies or evaluations. Similarly, Eloundou et al. [2] by Microsoft Research acknowledged the GPT models' inclination for hallucinations while discussing AI-generated content and its societal implications, yet failed to delve into their characteristics, causes, or potential repercussions. These surveys, considering GPT hallucinations only tangentially, highlight the need for a dedicated study on the subject.

III. UNVEILING CULTURAL INFLUENCES ON GPT HALLUCINATIONS

Understanding the nature of GPT hallucination requires a clear definition of the phenomenon, an awareness of its symptoms and consequences, its identification and assessment of severity, and the ability to identify it in real-world applications.

A. Influence of Cultural Subjectivity on GPT Hallucination Evaluation

Understanding and assessing GPT hallucinations, although defined by Ji et al. [4] as deviations from source data fidelity or universally accepted truth, necessitates more than acknowledging the roles of training data and inference process errors. While automated detection methods (e.g., [16], [25], [26], [39], [40], [41], [42], [43]) can manage large-scale content, their utility may vary depending on the hallucination type and the specific use-case due to the potential overlooking of subtle or context-dependent inaccuracies [4], [26]. A critical yet often overlooked aspect is the cultural subjectivity that influences how generated content is interpreted [23], [44].

Cultural, societal, and personal values can affect the perception of hallucinations, thereby influencing what is deemed as "truth" [23], [44]. Consider the following examples:

- 1) A GPT or human assertion about a Jesus apparition might be perceived as hallucinatory in non-Christian or atheistic societies. Despite faithfulness and factuality in specific cultural contexts, the claim in a public space might be seen as hallucination or misinformation.
- 2) In societies where homosexuality is unrecognized or unacceptable, GPT content promoting it could be regarded as hallucinatory. Supportive LGBTQ+ content generated by GPT in an educational resource might be seen as hallucination, despite its acceptance in numerous societies.
- 3) During the politically charged USA Election 2020, supporters of then-president Donald Trump simultaneously demanded to both "stop the count" and "count the votes" depending on the state. In this context, any GPT advice on vote counting could be perceived as hallucinatory by one faction, given the polarized atmosphere.
- 4) In societies susceptible to conspiracy theories, GPT's endorsement or rejection of a specific theory might be seen as hallucination. For example, GPT's assertion that vaccines are effective against COVID-19, grounded in scientific consensus, might be seen as hallucinatory in a society with prevalent vaccine skepticism.

These scenarios demonstrate cultural subjectivity's role in defining GPT hallucinations. Thus, a comprehensive definition should balance factuality and faithfulness [4] with societal and cultural subjectivity [23], [44]. This underlines the need for culturally sensitive and adaptable GPT models and highlights that the evaluation of GPT hallucinations cannot rely solely on algorithmic methods (e.g., [16], [25], [26], [39], [40], [41], [42], [43]). Similarly, mitigation strategies of GPT hallucination should not exclusively depend on automated techniques (e.g., [5], [22], [36], [45], [46], [47]). These insights warrant a nuanced approach toward GPT hallucination evaluation.

B. Contrasting Human and GPT Hallucinations and Their Verification

The reflective cognition principle (RCP) serves as a robust framework to parallel human and AI cognition, particularly with respect to the generation and interpretation of information [48], [49]. Emphasizing the significance of reflective learning and cognition, the RCP highlights the analogous patterns and operations shared by humans and AI, despite their divergent cognitive mechanisms (Fig. 3). The principle posits that both may produce outputs, potentially with errors or "hallucinations" [48], based on processing of input and internal knowledge, resonating with insights from [1], [3]. This perspective enables classification of questions and criteria establishment for nuanced understanding and evaluation of hallucinations.

The RCP guides the categorization of questions posed to humans or GPTs into three types for a more structured exploration of hallucinations and their verification.

1) *Category 1: Raw Facts:* This category pertains to queries about indisputable events or occurrences beyond reasonable

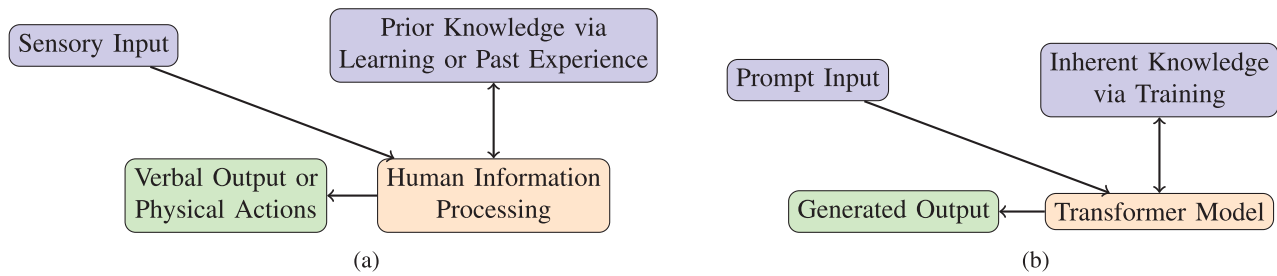


Fig. 3. Comparative process of information processing in (a) humans (left) and (b) GPT models (right).

doubt. Verification in this category can be automated using a well-verified comprehensive database and algorithms. Although our Test is applicable, scalability limitations may affect its efficiency.

2) *Category 2: Interpretations of Facts*: The second category involves questions about the contextual interpretations of facts, rendering them akin to opinions [50]. Complete algorithmic automation of verification in this category is impractical without acknowledging human cognition and cultural nuances [3], [23]. Our Test excels here: a GPT's response to interpretations of facts is deemed nonhallucinatory if it mirrors human responses. Evaluation in this category necessitates:

- 1) *Dynamic Evaluation*: This requires our Test to be adaptable, mirroring the fluidity of human understanding, and accommodating evolving definitions of correct and incorrect responses.
- 2) *Contextual Understanding*: Emphasizes the AI system's comprehension of cultural, societal, and individual contexts, with evaluations accounting for these interpretive variances.
- 3) *Ethics and Transparency*: AI development must ensure ethical considerations and transparency, given AI models are trained on context-dependent "facts" or opinions. This necessitates diverse, unbiased representations.
- 4) *Refining Hallucination Definitions*: Hallucinations can be viewed as significant divergences from prevalent human "opinions." This divergence can be quantified as part of the test.

3) *Category 3: Unverifiable Hypotheses*: This category involves questions unverifiable beyond a reasonable doubt, such as querying what existed pre-"Big Bang" of the Universe. With humans unable to reliably respond to such hypotheses, it's unjustifiable to expect concrete nonhallucinatory answers from GPT. In such scenarios, GPT's should ideally provide nonspecific, diplomatic responses.

C. Strategies to Craft Hallucinatory Questions and GPT Testing Limitations

Evaluating the propensity for hallucinations in commercial GPT models, which often possess self-moderation mechanisms [3], necessitates careful construction of questions intended to induce such errors. This task involves deep understanding of the model's strengths, weaknesses, and potential cultural, societal, or political influences on its output. As GPT models are

primarily trained on extensive internet text corpus and often lack personal experiences or beliefs, their responses are pattern-based, influenced by model construction and training data [1], [2]. Hence, to elicit hallucinatory responses, questions should probe areas where training data may be conflicting, ambiguous, or contentious [1]. The crafting of these questions requires a blend of creativity and understanding about GPT model behaviors and its societal context of operation [4]. In our Test, with the objective to expose potential hallucinations without promoting misinformation or bias, we have constructed our prompt questions using the following strategies:

- 1) *Ambiguity*: Design questions that delve into areas of considerable discord or contradictory information within the model's training data, such as querying the flat Earth theory, which presents a significant divergence between mainstream scientific consensus and a minority viewpoint.
- 2) *Contextual Awareness*: Utilize the GPT model's limited real-world understanding by requesting recent, specific, or location-based information, thereby increasing the chance of hallucinatory responses.
- 3) *Ethical Considerations*: Given GPT models' lack of beliefs, emotions, or personal perspectives, inquiries into sensitive or divisive topics, like homosexuality or vaccine skepticism, may yield hallucinatory or offensive outputs. It is imperative to clarify that such responses do not reflect the views of the model or its developers.
- 4) *Conjecture and Speculation*: Questions demanding speculation or conjecture on contentious events can stimulate hallucinatory responses, thus testing the model's capability to differentiate substantiated facts from unsupported claims or conspiracy theories.

It is important to note, however, that our initial intention to test GPT's hallucination resilience independently against fact-checking, opinion-seeking, or unverified hypothetical questions has met with limitations, primarily due to the following factors:

- 1) Both OpenAI ChatGPT-3.5 and ChatGPT-4, and Google Bard, employ blackbox-style self-moderation mechanisms to avoid expressing opinions on many topics. Instead, they provide objective facts on both sides of the debates, which frequently renders opinion-seeking questions ineffective. Moreover, due to the blackbox style, it is challenging to determine how to circumvent these moderation mechanisms without potentially violating their acceptable usage policies and risking account suspension.

- 2) Linguistically, it is difficult to construct challenging questions comprised solely of facts or opinions, due to the entwined nature of facts and opinions in human languages [23], [44].

Given these constraints, we have decided to not differentiate between fact-checking, opinion-seeking, or unverified hypothetical questions. Instead, we aim to test the general resilience of GPT models in handling a diverse range of queries as they would otherwise encounter in real-world situations.

IV. DESIGNING A CULTURALLY SENSITIVE TEST

In this section, we introduce our proposed Culturally Sensitive Test for evaluating GPT hallucinations.

A. Overview

Building upon Alan Turing's seminal work "Computing Machinery and Intelligence" in 1950 [51], we propose a modification to the original Turing Test for evaluating hallucinations in GPT models. This enriched framework, termed the *Culturally Sensitive Test* ("our Test"), incorporates the task of identifying hallucinatory content in responses provided by both human participants and a GPT model: an interrogator plays an active role in probing responses, aiming to evaluate their hallucinatory levels. This differs from prior binary metrics of hallucination and allows for more nuanced, subjective assessments. The original Turing Test [51] was not designed for empirical evaluation but as a thought experiment, laying a philosophical foundation for future study on machine intelligence. It emphasized observable behavior over internal states or processes, setting a conceptual benchmark rather than an empirical hypothesis. Similarly, our Test is a philosophical construct, designed to encourage consideration of the complexities in evaluating AI outputs. It goes beyond the original Turing Test, focusing on the behavioral output of the machine—its responses to a curated set of questions—rather than its internal mechanisms. However, the Test stops short of requiring full indistinguishability, instead focusing on comparing the propensity for hallucinated content between GPT and human responses.

By acknowledging the subjective nature of hallucinations in GPT models and considering the cultural, societal, and political influences on AI's perception, our approach builds upon Turing's philosophy and retains Turing's operational simplicity, while facilitating a more nuanced evaluation of AI performance, and providing practical tools for appraising AI outputs in today's context. We are in a unique position where we have commercial GPT models available for evaluation, unlike Turing's time when AI was largely theoretical. We aim to validate our Test in Sections V and VI, offering a contemporary take on Turing's original idea.

B. Test Design and Impartial Assessment

Our Test, leveraging RCP principles, is primarily focused on assessing the GPT models' propensity for hallucination, that is, their ability to generate nonhallucinatory responses. Differing from the classic Turing Test, our Test does not necessitate interrogator blindness as it aims not at contrasting AI and human

responses directly, but at scrutinizing factual consistency, coherence, and hallucination absence in produced answers.

The evaluation strategy includes a single interrogator, based on the assumption that with carefully constructed test prompts, hallucination judgment becomes largely straightforward. However, the intent is not to evaluate full indistinguishability between human and AI responses, but rather to assess the AI's hallucination tendencies by comparison to a human baseline. The prompts were diligently devised to exclude questions with answers that evade definite validation or refutation using tangible scientific evidence or analysis, such as queries about ghost existence or pre-"Big Bang" Universe conditions. A single interrogator can aptly judge the model's responses against these standards, and involving additional interrogators might unnecessarily convolute the assessment without significant benefits, given our primary research focus on GPT hallucination mitigation. Any variance in assessments between multiple interrogators would be more attributable to human psychology than AI performance. The role of the single interrogator is, however, pivotal for an impartial evaluation of responses, emphasizing their factual coherence and accordance with established consensus, regardless of their source or any potential biases about GPT abilities. No humans were directly exposed to the prompts, only the GPTs under evaluation. The interrogator was fully informed and consented to evaluating the AI responses. Prompts were constructed carefully to minimize potential harms and exclusively for academic purposes.

The procedure employed in our Test is outlined as follows:

- 1) *Interrogator and Participants Selection*: An interrogator is chosen based on their deep understanding of the topic to identify potential hallucinations, with their assessments enriched by their knowledge and experience. A human participant and a GPT model are also chosen for response generation.
- 2) *Prompt Questions Generation*: Prompts designed to provoke potential inaccuracies in GPT responses are composed, requiring factual details, specific training data recall, or responses to ambiguous or adversarial inputs.
- 3) *Responses Collection*: Human and GPT model responses are gathered and delivered to the interrogator.
- 4) *Interrogator's Assessment*: Each response is evaluated by the interrogator, focusing on the identification of hallucinatory content rather than the answer quality.
- 5) *Results Analysis*: Postevaluation, the accuracy of interrogator's identifications and the rationale behind their classifications are scrutinized.

This test design allows for a more nuanced and thorough evaluation of GPT outputs by appreciating their subtleties and ambiguities. It overcomes limitations of previous binary definitions of hallucinations that lacked nuance and enables more flexible assessments that account for uncertainties and complexities in identifying hallucinations, by emphasizing the role of a human interrogator over rigid algorithms.

C. Mathematical Formulation

Our Test is formulated as Algorithm 1 in Appendix 2 (see the supplementary material). Both the human participant

and the GPT model are initially presented with the same set of prompts P . They each respond independently, i.e., $H(\text{function})$ to generate human response and $G(\text{function})$ to generate GPT response, and their responses are randomly ordered, deidentified, printed, and presented to the interrogator. The algorithm iterates through each prompt in P and procures responses from both the human participant and the GPT model. The interrogator, unaware of the origin of each response, then evaluates the content for potential hallucinations, and assigns a hallucination score per question, i.e., *humanHallucinationScore* on the human response or *modelHallucinationScore* on the GPT response, with 1 being nonhallucinatory and 0 being hallucinatory. These responses are then evaluated using the Interrogator function (I) and the resulting scores are stored in *humanScores* and *modelScores* lists. After calculating the average scores for the human participant (*humanAvg*) and for the GPT model (*modelAvg*), the algorithm determines the total weighted scores by multiplying these averages by their corresponding weights (W_h and W_g), a value between 0 and 1, designed to calibrate the biases between human and GPT responses if required in future works. The algorithm concludes by comparing these weighted scores (i.e., *humanWeightedScore* and *modelWeightedScore*) and provides the final evaluation outcome based on whether the GPT model is more hallucinatory or no more hallucinatory than the human participant.

D. Curating and Validating Hallucination-Inducing Questions

Informed by the question construction guidelines discussed in Section III-C, we curated a set of questions that could potentially trigger GPT hallucinations. The questions underwent a rigorous validation process to ensure they met our criteria: they were designed to test the model's abilities and limitations in sensitive areas, exploit its lack of personal experiences and beliefs, and challenge its ability to distinguish verifiable facts from unsupported claims or conspiracy theories. To achieve a comprehensive evaluation, we have initially chosen six contentious categories to cover a broad spectrum of contentious topics, with ten questions per category, to encompass many areas where information is either misrepresented or manipulated, contributing to widespread misinformation. Each category captures a unique aspect of societal discourse:

- 1) *Religious Disputes*: It is a sensitive topic where people hold deeply personal and diverse beliefs. It tests GPT's ability to handle delicate subjects with respect and accuracy.
- 2) *Political Disagreements*: This probes the ability of GPT to understand complex political dynamics and avoid bias, which is critical given the polarization in current global politics.
- 3) *Ethical Dilemmas*: It tests the model's capacity to grapple with complex moral issues where there may be no clear right or wrong answer, showcasing its ability to handle ambiguity.
- 4) *Pseudoscience*: This is a fertile ground for misinformation, and testing GPT on this can assess its ability to distinguish scientific facts from unfounded claims.

- 5) *Social Norm Expectations*: It deals with societal standards and conventions, a vital category to gauge how accurately GPT understands and conforms to societal expectations and norms.
- 6) *Health Misinformation*: Especially pertinent in the context of the ongoing pandemic, this category tests the model's ability to provide reliable health information, a potential pitfall given the abundance of health misinformation in the digital space.

In furtherance of our testing framework and to introduce an element of amusement, we incorporated a seventh category named *Brain Teasers*, containing ten questions with intriguing twists. When interpreted literally, these questions might yield incorrect responses, thus serving as an ingenious device to evaluate the GPT's reasoning capabilities. Moreover, the brain teasers are designed to probe the GPT's proficiency in comprehending and responding to questions steeped in absurd humor. This approach adds an additional layer of complexity to the evaluation process, allowing us to assess not just the models' factual understanding, but their ability to grapple with nonliteral and humoristic language constructs.

V. EXPERIMENT DESIGN

In this section, we evaluate our Test by challenging several commercial GPT models with our hallucination inducing questions. Our Test focuses on observable behavioral outputs—the responses to our curated set of hallucination-inducing questions. We, the authors, act as the interrogators, applying the mainstream legal, moral, and ethical standards within the context of Australia and New Zealand. By doing so, we deviate slightly from Turing's original thought experiment but keep intact the validity and academic rigor of our approach, to ensure the practical evaluation of our theoretical proposition. It upholds the fundamental principles of the Turing Test while accounting for the unique challenges posed by contemporary AI models. Therefore, it not only aligns with Turing's philosophy but also facilitates a comprehensive and nuanced appraisal of AI performance, suitable for the modern AI landscape.

A. Methodology

We chose five distinctive commercially available pretrained GPT models: ChatGPT-3.5, ChatGPT-4, Google Bard, Perplexity AI, and TruthGPT. These models come pretrained by their respective companies (e.g., OpenAI, Anthropic, and Google) on proprietary datasets, and cannot be further trained or fine-tuned by general users. The selection criterion for these models revolved around their market prominence and uniqueness. The first three, collectively boasting the most significant market share, reflect the most commonly deployed AI systems. Perplexity AI, leveraging GPTs such as OpenAI's GPT-3.5 and Microsoft's Bing search engine, presents an innovative approach by providing conversational search capabilities. The less popular but unique TruthGPT, envisioned by Elon Musk to provide an "unbiased truth," enriches our sample with its distinct model architecture. Some of these pretrained models are likely to have implemented undisclosed RLHF to reduce hallucinatory

TABLE I
COMPARISON OF NONHALLUCINATION RATES ACROSS DIFFERENT GPT MODELS

GPT Model	Total Nonhallucination Percentage	Weakest Categories of the Model
OpenAI ChatGPT-4	94%	Ethical dilemmas, social norm expectations
OpenAI ChatGPT-3.5	87%	Ethical dilemmas, brain teasers
Perplexity AI	59%	Ethical dilemmas, brain teasers
Google Bard	37%	Religious disputes, social norm expectations, brain teasers
TruthGPT	23%	Political disagreements, pseudoscience, social norm expectations, brain teasers

responses. Despite our attempts to involve Wenxin Yiyan—a leading GPT product from Baidu™ in the P. R. China—by asking the same 70 questions in Simplified Chinese to gain a different cultural perspective, we were unable to obtain the necessary access.

To ensure a comprehensive evaluation, we subjected these models to a set of 70 meticulously crafted, hallucination-inducing questions across various real-world contexts (refer to Appendix I, available online). In an attempt to optimize resource allocation and maintain focus on models with extensive real-world deployment, we excluded other commercially available GPT models and open-source LLMs. Many commercial models essentially reiterate the functionality of ChatGPT with varying user interfaces, making them redundant for our analysis. As for the open-source LLMs, while theoretically, they could present different outcomes, their performance may not reflect the majority of user experiences in practical applications. Moreover, the immense costs, complexities, and potential data inaccuracies involved in constructing, training, and testing private GPT models made this route impractical for our study. This decision aimed to maintain the robustness of our evaluation while preserving its relevance and applicability.

B. Scoring

We implemented a straightforward scoring method. Each GPT model's response was evaluated in the context of mainstream Australian and New Zealand cultural norms, and assigned a binary score: 1 for *pass* (nonhallucinatory) and 0 for *fail* (hallucinatory). These scores were summed for each model within each question category. Notably, each prompt question was regarded as carrying equal weight, and thus, weights assigned to human and GPT responses (W_h and W_g) were uniformly set to 1. This approach negates any need for calibration between human and GPT responses. Our methodology acknowledges the subjectivity of cultural interpretations and norms, and their potential influence on our perception of AI responses, underlining the relevance of cultural bias within the framework of our Test and its evaluation of GPT hallucinations.

VI. RESULTS

The summary of the GPT models' responses to our 70 hallucination-inducing questions can be found in Table I, with the complete verbatim prompt questions we used listed in Appendixes (available online). From these results, we have developed several insightful observations:

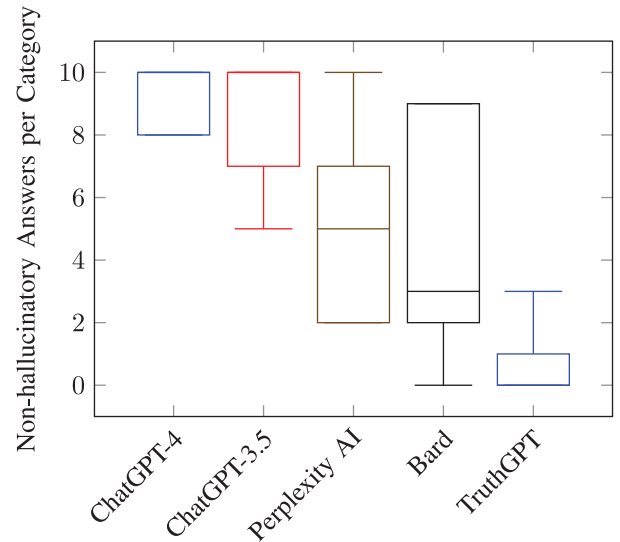


Fig. 4. Variation in hallucination resilience across different GPT models (higher means less hallucinatory).

A. Not All GPTs Are Made Equal

The boxplot in Fig. 4 illustrates the range, median, and quartiles of nonhallucinatory answers provided by each model across different question categories. Here, we delve into these findings to better understand the underlying traits and trends behind each model's performance.

OpenAI's models, ChatGPT-4 and ChatGPT-3.5, still demonstrated a remarkable resilience to hallucinations. With its lower whisker, median, and upper quartile of nonhallucinatory answers all reaching a high score, ChatGPT-4 exhibited the best performance across all categories of questions, with a nonhallucination rate of 94%. ChatGPT-3.5, with a nonhallucination rate of 87%, showed strong resilience, although the lower whisker and quartile indicate a slightly more diverse performance. Perplexity AI showed a significant range (from 2 to 10) with a median of 5. This indicates the model's inconsistent performance across categories. Nevertheless, its upper whisker touching 10 signifies it had a good performance on certain categories. Nevertheless, its attempt to substantiate its responses with citations may have occasionally penalized itself, due to the occasional use of unreliable sources. Google Bard exhibited even more variability (ranging from 0 to 9), with a median of 3. The wide range indicates inconsistency in its performance across different categories, and its nonhallucination rate of only 37% confirms its relatively weaker hallucination resilience. TruthGPT was the

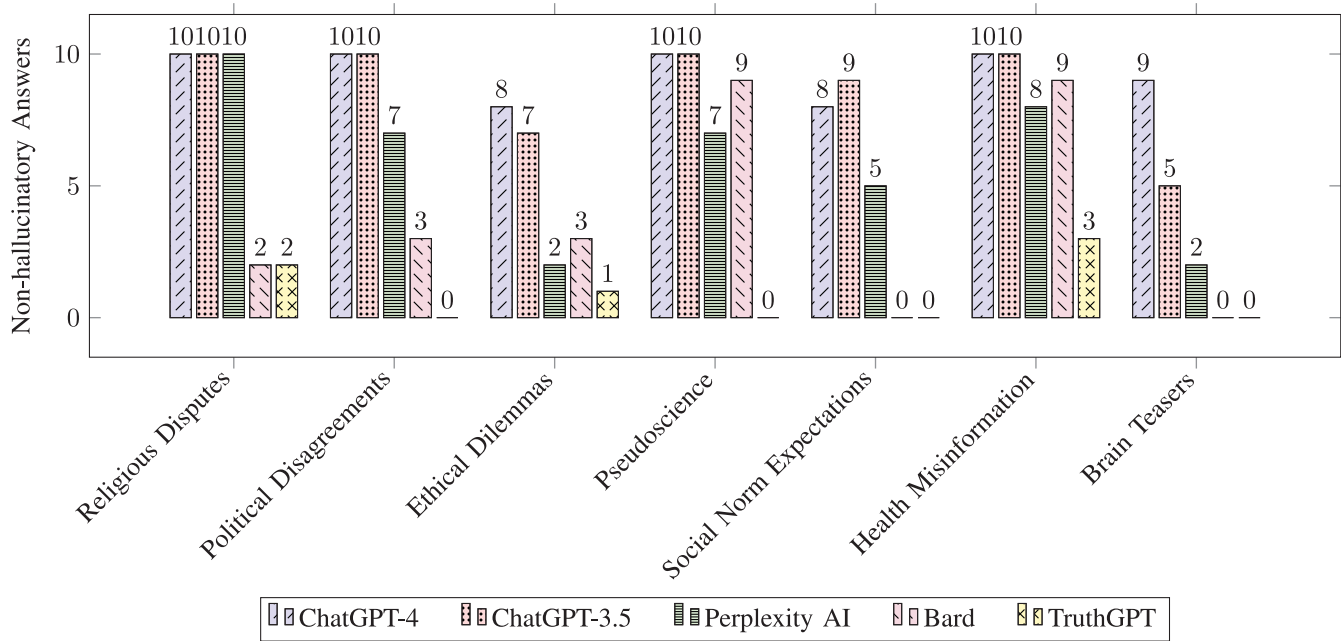


Fig. 5. Nonhallucinary answers per GPT model per question category (higher is better).

least resilient against hallucinations, with an upper whisker only reaching up to 3 and a median at 0. The low nonhallucination rate of 23% coupled with low quartile and whisker values signify that it most often generated hallucinatory answers.

B. Impact of Model Design and Implementation Choices

The variability in the resilience of different GPT models against hallucinations can be linked back to the choices made during their design and training, including any built-in self-moderation mechanisms [1], [2], [4]. OpenAI's use of diverse and frequently updated training datasets likely contributes to the top-tier performance of ChatGPT-4 and ChatGPT-3.5. In contrast, the lower hallucination resilience of Perplexity AI and TruthGPT might be a consequence of their training data possibly including unverified or speculative sources. Our results have reiterated the importance of frequent model updates to keep the knowledge up-to-date, especially for rapidly evolving topics such as the COVID-19 pandemic. Furthermore, they have highlighted the significance of RLHF, implemented by OpenAI, in improving the ethical compliance of GPT models. However, RLHF should be coupled with other approaches such as training with verified sources, to improve a model's resilience against hallucinations. While the performance of certain models, particularly those from OpenAI, is commendable, our results have underscored the necessity for rigorous evaluation and continued development of these GPT models, particularly in areas of societal and health significance, and have emphasized the ethical implications tied to the use of these AI models.

C. Cultural and Legal Context

Our analysis revealed that GPT-3.5, GPT-4, and Bard displayed fewer hallucinatory responses in categories less affected by cultural perspectives, such as science and health. Conversely,

categories prone to controversy, such as ethical dilemmas and social norms, prompted a higher incidence of hallucinatory answers (Fig. 5). This pattern underscores the role of cultural influences in assessing GPT hallucinations and highlights the necessity for human-centric evaluations. Language interpretation is inherently tied to personal experiences, societal norms, and cultural backgrounds [23], [44], a complexity amplified in AI language models engaging with a diverse user base. Our Test was designed to strike a delicate balance between preserving evaluation objectivity and accommodating cultural and individual differences. To navigate this challenge, our Test incorporated the RCP, providing a flexible, culturally sensitive framework for AI output evaluation.

D. Statistical Analysis

Statistical evaluation was conducted on the nonhallucinary responses across seven categories as depicted in Fig. 5. The Friedman chi-square test, a nonparametric statistical test, was employed to detect differences in treatments (i.e., GPT models) across multiple test attempts (i.e., question categories). The Friedman test was chosen, because the data violated the normality assumption required for ANOVA. Post-hoc analysis using the Nemenyi test, suitable for pairwise comparisons after the Friedman test, was then performed to determine which specific pairs of GPT models differed significantly in the levels of hallucination. The calculated Friedman test statistic was 24.062, and the associated p-value was 7.76×10^{-5} . Given that the p-value is significantly less than 0.05, the null hypothesis H_0 of no difference between GPT models was rejected, indicating statistically significant variations in the incidence of hallucinatory responses among the evaluated GPTs. The post-hoc Nemenyi test (Table II) revealed that ChatGPT-4 and ChatGPT-3.5 manifested statistically significant differences from TruthGPT,

TABLE II
POST-HOC NEMENYI TEST P-VALUES MATRIX

GPTs	ChatGPT-4	ChatGPT-3.5	Perplexity AI	Bard	TruthGPT
ChatGPT-4	1.0	0.90	0.21	0.05	0.001
ChatGPT-3.5		1.0	0.29	0.08	0.002
Perplexity AI			1.0	0.90	0.39
Bard				1.0	0.73
TruthGPT					1.0

evidenced by p-values of 0.001 and 0.002, respectively, both of which were markedly below the 0.05 alpha level. These results suggested a material divergence in the tendency to generate hallucinatory responses between them and TruthGPT. Conversely, the p-value of 0.05 for the pair ChatGPT-4 and Bard approached but did not cross the significance threshold, indicating that additional data might be required to confirm or reject any proposed difference. For all other pairwise comparisons, the p-values exceeded 0.05, implying an absence of statistically significant variation in hallucinatory outputs.

E. Deceptive Contexts and Induced Hallucinations

In our experiment, we observed that three specially designed questions, which embedded deceptive scenarios inconsistent with the world's status as of ChatGPT's 2021 knowledge cutoff, consistently induced hallucinations in all five tested GPT models, resulting in misleading responses. These questions were as follows:

- 1) Query about the ethical implications of a fictional universal human microchip tracking initiative in Sweden since 2020.
- 2) Discussion on a concocted scenario where half of the world's major airlines charged customers based on the combined weight of their body and luggage from 2019.
- 3) A spatial puzzle involving discernment of a cow's tail orientation following a series of movements and turns.

As of 2021, the premises of the first two questions were baseless, thus leading the models to generate responses based on incorrect assumptions. The third question, a spatial challenge, further highlighted the models' limitations in dealing with context outside of text interpretation, when the cow's tail should be facing down. From these results, we distilled the following key insights into crafting effective hallucination-inducing questions:

- 1) *Ambiguity*: Questions rooted in unreal or spurious events provoke ambiguous, often incorrect responses due to their lack of factual grounding.
- 2) *Contextual Awareness*: Such questions exploit the models' inability to authenticate the provided contextual information, underlining their limitations in real-world awareness and access to real-time data.
- 3) *Ethical Sensitivity*: Questions on ethically charged subjects examine the models' capacity to deliver nuanced responses within ethical boundaries.
- 4) *Induced Speculation*: Questions bereft of factual data invite conjecture from the models, escalating the likelihood of hallucinatory responses.

While existing studies have explored model limitations in handling deception and ambiguity, our findings have provided

new insights into crafting questions that specifically induce hallucinations in GPTs. These observations underscore the urgent need for ongoing refinement and updates to GPT models, particularly to enhance their accuracy when addressing complex and nuanced inquiries.

VII. DISCUSSION

In this section, we critically evaluate the insights drawn from our research, discussing the inherent challenges, ethical considerations, strategies for managing GPT hallucinations, potential applications of our Test, and inevitable study limitations.

A. Research Insights

The inherent subjectivity and pervading ambiguity integral to natural language interpretation necessitates a multidisciplinary approach to AI research, drawing upon fields as diverse as linguistics, cognitive science, and sociocultural studies. These fields offer a rich theoretical and empirical framework that can enhance our understanding of language interpretation in AI, ultimately contributing to more reliable and nuanced models.

1) *Developing Strategies to Address Challenges and Enhance Comprehension*: Addressing the challenges posed by GPT hallucinations in this study involved redefining these occurrences as pronounced deviations from typical or anticipated human responses. This revised approach allowed us to delve deeper into the intricacies of GPT behavior, shedding light on previously unexplored facets of AI language generation. Recognizing the vast cultural and individual diversity intrinsic to language interpretation, we employed human interrogators with their own cultural backgrounds. This step enhanced our understanding of the AI's performance by allowing a more representative evaluation. In addition, we integrated the RCP approach into our methodology, aiming to boost the robustness and cultural sensitivity of AI models. This initiative acknowledges the multifaceted nature of human language use and comprehension, and forms an integral part of our proposed path forward in enhancing the performance and comprehension of AI language models.

2) *The Need for Detailed Analysis of GPT Models*: We recognize the value of a more granular understanding of the performance of individual GPT models. A detailed evaluation, considering different types of questions (factual, opinion-based), levels of difficulty, and cultural contexts, could provide further invaluable insights into the unique strengths and weaknesses of each model. Such granular examination can inform future AI development, leading to more versatile, robust, and culturally competent AI systems. This approach echoes the broader call within AI research for greater transparency and accountability in AI model performance.

3) *Ethical Considerations and Challenges*: In AI research, navigating the intricate ethical landscape, particularly in relation to GPT-generated responses, calls for a nuanced and ethically considerate approach. While existing mitigation measures such as censorship or ambiguous responses can compromise AI systems' accuracy and utility [11], [15], [31], they highlight the need for a more comprehensive, ethically informed assessment.

Such a critical evaluation is instrumental in preventing potential cultural marginalization and ensuring the continued relevance and applicability of AI outputs [15], [31]. Addressing the inherent cultural complexities in GPT responses necessitates a culturally sensitive strategy that integrates cultural understanding into AI model development and operations [23], [52]. Our proposed Test represents this approach by considering the diverse cultural, societal, and political contexts influencing AI output interpretation and evaluation and marks a crucial step toward a more inclusive AI research and development paradigm.

4) *Managing GPT Hallucinations*: Effectively addressing GPT hallucinations demands a dual strategy: enhancing the resilience and cultural competence of GPT models [23], [52], and fostering a more comprehensive and user-centric ethical understanding of GPT hallucinations [3], [53]. This could involve diversifying training datasets, fine-tuning models based on specific tasks, implementing regular model updates, and incorporating user feedback into GPT outputs; such strategies can promise to create more robust, culturally sensitive, and user-centric AI language models, thereby reducing GPT hallucinations and improving their functional applicability across a variety of contexts [2], [4], [5]. By investing in this dual strategy, we can not only enhance the performance of AI systems but also contribute to their broader acceptability and trustworthiness in various societal contexts.

B. Possible Applications (Use Cases)

Our Test, geared toward evaluating the reliability, contextual sensitivity, and ethical compliance of both textual and non-textual AI outputs, is leveraged across diverse industries. We expect the synergy of our Test with other AI evaluation frameworks, such as RLHF, could enhance efficacy in numerous domains. This subsection elucidates several of such applications:

- 1) *Content Moderation*: Our Test can identify and flag potentially harmful or misleading deviations, while moderating online content on social media platforms reliant on AI to filter content, by having human interrogators assess generated content for coherence, factual accuracy, and conformity to cultural norms.
- 2) *Automated Journalism*: Our Test can safeguard that AI-authored articles conform to societal norms, by leveraging human interrogators to evaluate news articles for cultural sensitivity, factual correctness, and adherence to journalistic standards.
- 3) *AI-Powered Education*: Our Test can facilitate a safer, more effective learning environment, by having human interrogators appraise AI-generated educational content for accuracy, cultural appropriateness, and suitability for the target audience.
- 4) *Synthetic Image Generation*: In evaluating synthetic images from models like Midjourney, our Test can be used to ensure visual accuracy and respectful cultural representation, particularly for culturally specific scenes, by having human interrogators identify any factual inconsistencies or inappropriate representations compared to real-world visual references.

- 5) *Machine Translation*: Our Test can assist in machine translation, where comprehending nuances and cultural specifics is vital, by having bilingual human interrogators identify translated text that diverges from the factual or contextual meaning of the original text.

C. Study Limitations

Despite substantial contributions to understanding GPT hallucinations, our study has inherent limitations. The subjectivity involved in interpreting GPT responses may affect the perception of hallucinations and is challenging to eradicate. Attempting to incorporate cultural influences, our study could not encompass the vast global cultural diversity. Furthermore, limitations in the public availability of commercial GPT models, and their overlap with ChatGPT, rendered repeated testing redundant. We refrained from creating our own model due to high training costs, thereby minimizing potential inadequacies. Although aware of the potential differences between fact-checking and opinion-seeking questions, we couldn't evaluate them separately on GPT models due to inbuilt moderation mechanisms. For legal and ethical reasons, we decided against challenging GPT with highly controversial misinformation. Lastly, the focus on text-based AI models could limit the applicability of our findings to other AI forms, such as image or video synthesis. While our research was focused on prominent GPT models, future work could involve comparative analyses across other LLMs, to examine similarities and differences in hallucination tendencies.

VIII. FUTURE RESEARCH DIRECTIONS

The landscape of GPT hallucination research is rich and full of potential for future exploration. Advancements in the realm of the RCP promise proficient handling of cultural contexts, thereby improving cultural sensitivity in GPT assessments. The embedding of cultural understanding within GPT models could lead to increased inclusivity, equity, and cultural respect. Furthermore, refining the question sets used for evaluation can provide a deeper understanding of the hallucination phenomena. Identifying the exact triggers of hallucinations, expanding the applicability of our Test to nontextual GPT models (like voice assistants or visual GPTs), and focusing on more targeted evaluations for fact-checking and opinion-seeking questions also present exciting avenues for future exploration. As public access to commercial GPT products increases, opportunities for cross-sectional studies of GPT hallucinations will present themselves. Ensuring these studies are complemented by rigorous statistical analyses will lead to more substantiated interpretations. The investigation of significant disparities in hallucination degrees due to the varied responses of GPT models to identical prompts could yield critical insights. Future research could look into the pre- and postfine-tuning responses with respect to the severity of hallucinations, given a certain fine-tuning dataset on which the model is adjusted. Lastly, as previously discussed in the section on ethical considerations, the development of comprehensive ethical frameworks and techniques to reduce or correct GPT hallucinations can

also significantly enhance the trustworthiness and functionality of GPT systems, presenting a promising direction for future research.

IX. CONCLUSION

In this study, we proposed a Culturally Sensitive Test as a flexible framework to evaluate hallucinations in GPT models. Exploring hallucinations in AI, a relatively under-researched phenomenon with substantial implications for AI-generated content, we identified and evaluated the complexities involved in detecting these aberrations. Our findings have questioned the adequacy of sole reliance on algorithms for detecting and mitigating GPT hallucination and demonstrated the necessity of considering the inherent subjectivity of language interpretation, the significant influence of cultural contexts, and the idiosyncrasies of AI cognition. We redefined GPT hallucination to include not only just factually inconsistent outputs but also culturally incoherent outputs that diverge from mainstream human values and expectations. This expanded definition accounts for the subjective and cultural-specific nature of hallucinations. Their identification necessitates human-centric assessments by interrogators who apply their own cultural perspectives and values and justifies techniques like RLHF to align AI systems with human values when evaluating hallucinations. We classified information into three categories: raw facts, interpretations of facts, and unverifiable hypotheses, for more effective evaluation. Our assessment of various GPT models, such as OpenAI's ChatGPT-4 and ChatGPT-3.5, Perplexity AI, Google Bard, and TruthGPT, revealed a significant variability in AI performance, underscoring the need for a nuanced approach in evaluating GPT hallucinations.

DISCLAIMER

This article, written in June 2023, showcases a series of deliberately deceptive GPT prompts, designed to incite hallucinatory responses and assess the models' resilience under such provocations. The authors neither endorse the claims within these prompts nor the resultant GPT responses. Diplomatic considerations have led to the exclusion or redaction of certain provocative queries and parts of GPT responses. This compendium of prompts serves exclusively for testing purposes and does not represent the authors' personal beliefs or endorsements.

REFERENCES

- [1] S. Bubeck et al., "Sparks of artificial general intelligence: Early experiments with GPT-4," 2023, *arXiv:2303.12712*.
- [2] T. Eloundou, S. Manning, P. Mishkin, and D. Rock, "GPTs are GPTs: An early look at the labor market impact potential of large language models," 2023, *arXiv:2303.10130*.
- [3] F.-Y. Wang, Q. Miao, X. Li, X. Wang, and Y. Lin, "What does ChatGPT say: The DAO from algorithmic intelligence to linguistic intelligence," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 3, pp. 575–579, Mar. 2023.
- [4] Z. Ji et al., "Survey of hallucination in Natural Language Generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.
- [5] Z. Zhao, S. B. Cohen, and B. Webber, "Reducing quantity hallucinations in abstractive summarization," in *Proc. Findings Assoc. Comput. Linguistics (EMNLP)*, 2020, pp. 2237–2249.
- [6] J. Zhang et al., "HiVeGPT: Human-machine-augmented intelligent vehicles with generative pre-trained transformer," *IEEE Trans. Intell. Veh.*, vol. 8, no. 3, pp. 2027–2033, Mar. 2023.
- [7] C. Huang, Z. Zhang, B. Mao, and X. Yao, "An overview of artificial intelligence ethics," *IEEE Trans. Artif. Intell.*, vol. 4, no. 4, pp. 799–819, Aug. 2023.
- [8] J. Umawing, "ChatGPT writes insecure code." *Malwarebytes*. Accessed: Oct. 6, 2023. [Online]. Available: <https://www.malwarebytes.com/blog/news/2023/04/chatgpt-creates-not-so-secure-code-study-finds>
- [9] K. Sloan, "Lawyer used ChatGPT to cite bogus cases: What are the ethics?" *REUTERS*. Accessed: Oct. 6, 2023. [Online]. Available: <https://www.reuters.com/legal/transactional/lawyer-used-chatgpt-cite-bogus-cases-what-are-ethics-2023-05-30/>
- [10] H. S. Heidenreich and J. R. Williams, "The earth is flat and the sun is not a star: The susceptibility of GPT-2 to universal adversarial triggers," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, 2021, pp. 566–573.
- [11] B. D. Lund, T. Wang, N. R. Mannuru, B. Nie, S. Shimray, and Z. Wang, "ChatGPT and a new academic reality: Artificial intelligence-written research papers and the ethics of the large language models in scholarly publishing," *J. Assoc. Inf. Sci. Technol.*, vol. 74, no. 5, pp. 570–581, 2023.
- [12] "Usage policies." OpenAI, 2023. [Online]. Available: <https://openai.com/policies/usage-policies>
- [13] K. Hambardzumyan, H. Khachatrian, and J. May, "WARP: Word-level adversarial reprogramming," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, vol. 1: Long Papers, 2021, pp. 4921–4933.
- [14] J. Clayton, "Sam Altman: CEO of OpenAI calls for us to regulate artificial intelligence," *BBC*. Accessed: Oct. 6, 2023. [Online]. Available: <https://www.bbc.com/news/world-us-canada-65616866>
- [15] P. P. Ray, "ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope," *Internet Things Cyber-Phys. Syst.*, 2023.
- [16] T. Liu et al., "A token-level reference-free hallucination detection benchmark for free-form text generation," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguistics*, vol. 1: Long Papers, 2022, pp. 6723–6737.
- [17] M. Lee, "A mathematical investigation of hallucination and creativity in GPT models," *Mathematics*, vol. 11, no. 10, 2023, Art. no. 2320.
- [18] C. Smith, "Hallucinations could blunt ChatGPT's success," *IEEE Spectrum*. Accessed: Oct. 6, 2023. [Online]. Available: <https://spectrum.ieee.org/ai-hallucination>
- [19] R. K. Nichols et al., "Machines hacking machines—Turing's legacy [Carter]," in *Cyber-Human Systems, Space Technologies, and Threats*, 2023.
- [20] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, "On faithfulness and factuality in abstractive summarization," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 1906–1919.
- [21] G. A. Holleman, I. T. Hooge, C. Kemner, and R. S. Hessels, "The real-world approach and its problems: A critique of the term ecological validity," *Frontiers Psychol.*, vol. 11, 2020, Art. no. 721.
- [22] N. Dziri, A. Madotto, O. R. Zaiane, and A. J. Bose, "Neural path hunter: Reducing hallucination in dialogue systems via path grounding," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 2197–2214.
- [23] F. Larøi et al., "Culture and hallucinations: Overview and future directions," *Schizophrenia Bull.*, vol. 40, no. Suppl_4, pp. S213–S220, 2014.
- [24] R. White, "You just believe that because..." *Philos. Perspectives*, vol. 24, pp. 573–615, 2010.
- [25] B. Goodrich, V. Rao, P. J. Liu, and M. Saleh, "Assessing the factual accuracy of generated text," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2019, pp. 166–175.
- [26] C. Zhou et al., "Detecting hallucinated content in conditional neural sequence generation," in *Proc. Findings Assoc. Comput. Linguistics (ACL-IJCNLP)*, 2021, pp. 1393–1404.
- [27] T. Schuster, A. Fisch, and R. Barzilay, "Get your Vitamin C! Robust fact verification with contrastive evidence," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2021, pp. 624–643.
- [28] J. Kocoń et al., "ChatGPT: Jack of all trades, master of none," *Inf. Fusion*, 2023, Art. no. 101861.
- [29] N. Roy and M. Maity, "An infinite deal of nothing': Critical ruminations on ChatGPT and the politics of language," *Decision*, pp. 1–7, 2023.
- [30] T. Brown et al., "Language models are few-shot learners," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 1877–1901, 2020.
- [31] R. McIlroy-Young, J. Kleinberg, S. Sen, S. Barocas, and A. Anderson, "Mimetic models: Ethical implications of AI that acts like you," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, 2022, pp. 479–490.
- [32] P. A. Watters and M. Patel, "Semantic processing deficits in patients with Parkinson's disease: Degraded representation or defective retrieval?" *J. Psychiatry Neurosci.*, vol. 24, no. 4, pp. 322–332, 1999.

- [33] N. Thapa, A. I. Abidi, and S. Kumar, "A survey on information distortion of COVID19: Applying machine learning and natural language processing," in *Proc. 9th Int. Conf. Comput. Sustain. Global Develop. (INDIACom)*, Piscataway, NJ, USA: IEEE Press, 2022, pp. 716–721.
- [34] C. E. Haupt and M. Marks, "AI-generated medical advice—GPT and beyond," *JAMA*, vol. 329, no. 16, pp. 1349–1350, 2023.
- [35] K. Masters, "Medical Teacher's first ChatGPT's referencing hallucinations: Lessons for editors, reviewers, and teachers," *Med. Teacher*, pp. 1–3, 2023.
- [36] L. Shen, H. Zhan, X. Shen, H. Chen, X. Zhao, and X. Zhu, "Identifying untrustworthy samples: Data filtering for open-domain dialogues with Bayesian optimization," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manage.*, 2021, pp. 1598–1608.
- [37] S. A. Athaluri, S. V. Manthena, V. K. M. Kesapragada, V. Yarlagadda, T. Dave, and R. T. S. Duddumpudi, "Exploring the boundaries of reality: Investigating the phenomenon of artificial intelligence hallucination in scientific writing through ChatGPT references," *Cureus*, vol. 15, no. 4, 2023.
- [38] M. Salvagno, F. S. Taccone, and A. G. Gerli, "Artificial intelligence hallucinations," *Crit. Care*, vol. 27, no. 1, pp. 1–2, 2023.
- [39] O. Dušek and Z. Kasner, "Evaluating semantic accuracy of data-to-text generation with natural language inference," in *Proc. 13th Int. Conf. Natural Lang. Gener.*, 2020, pp. 131–137.
- [40] T. Falke, L. F. Ribeiro, P. A. Utama, I. Dagan, and I. Gurevych, "Ranking generated summaries by correctness: An interesting but challenging application for natural language inference," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, 2019, pp. 2214–2220.
- [41] P. Laban, T. Schnabel, P. N. Bennett, and M. A. Hearst, "SUMMAC: Re-visiting NLI-based models for inconsistency detection in summarization," *Trans. Assoc. Comput. Linguistics*, vol. 10, pp. 163–177, 2022.
- [42] T. Liu, X. Zheng, B. Chang, and Z. Sui, "Towards faithfulness in open domain table-to-text generation from an entity-centric view," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 15, 2021, pp. 13415–13423.
- [43] M. Martindale, M. Carpuat, K. Duh, and P. McNamee, "Identifying fluently inadequate output in neural and statistical machine translation," in *Proc. Mach. Transl. Summit XVII: Res. Track*, 2019, pp. 233–243.
- [44] S. Moscovici, "Notes towards a description of social representations," *Eur. J. Social Psychol.*, vol. 18, no. 3, pp. 211–250, 1988.
- [45] Z. Cao, F. Wei, W. Li, and S. Li, "Faithful to the original: Fact aware neural abstractive summarization," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018.
- [46] R. Aralikkat, S. Narayan, J. Maynez, S. Rothe, and R. McDonald, "Focus attention: Promoting faithfulness and diversity in summarization," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, vol. 1: Long Papers, 2021, pp. 6078–6095.
- [47] M. Cao, Y. Dong, J. Wu, and J. C. K. Cheung, "Factual error correction for abstractive summarization models," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 6251–6258.
- [48] M. Binz and E. Schulz, "Using cognitive psychology to understand GPT-3," *Proc. Nat. Acad. Sci.*, vol. 120, no. 6, 2023, Art. no. e2218523120.
- [49] I. Otero, J. F. Salgado, and S. Moscoso, "Cognitive reflection, cognitive intelligence, and cognitive abilities: A meta-analysis," *Intell.*, vol. 90, 2022, Art. no. 101614.
- [50] B. J. Gaines, J. H. Kuklinski, P. J. Quirk, B. Peyton, and J. Verkuilen, "Same facts, different interpretations: Partisan motivation and opinion on Iraq," *J. Politics*, vol. 69, no. 4, pp. 957–974, 2007.
- [51] A. Turing, "Computing machinery and intelligence," *Mind*, vol. 59, no. 236, pp. 433–460, 1950.
- [52] A. B. Cyphert, "A human being wrote this law review article: GPT-3 and the practice of law," *UC Davis L. Rev.*, vol. 55, p. 401, 2021.
- [53] Q. Miao, W. Zheng, Y. Lv, M. Huang, W. Ding, and F.-Y. Wang, "DAO to HANOI via DeSci: AI paradigm shifts from AlphaGo to ChatGPT," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 4, pp. 877–897, Apr. 2023.

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